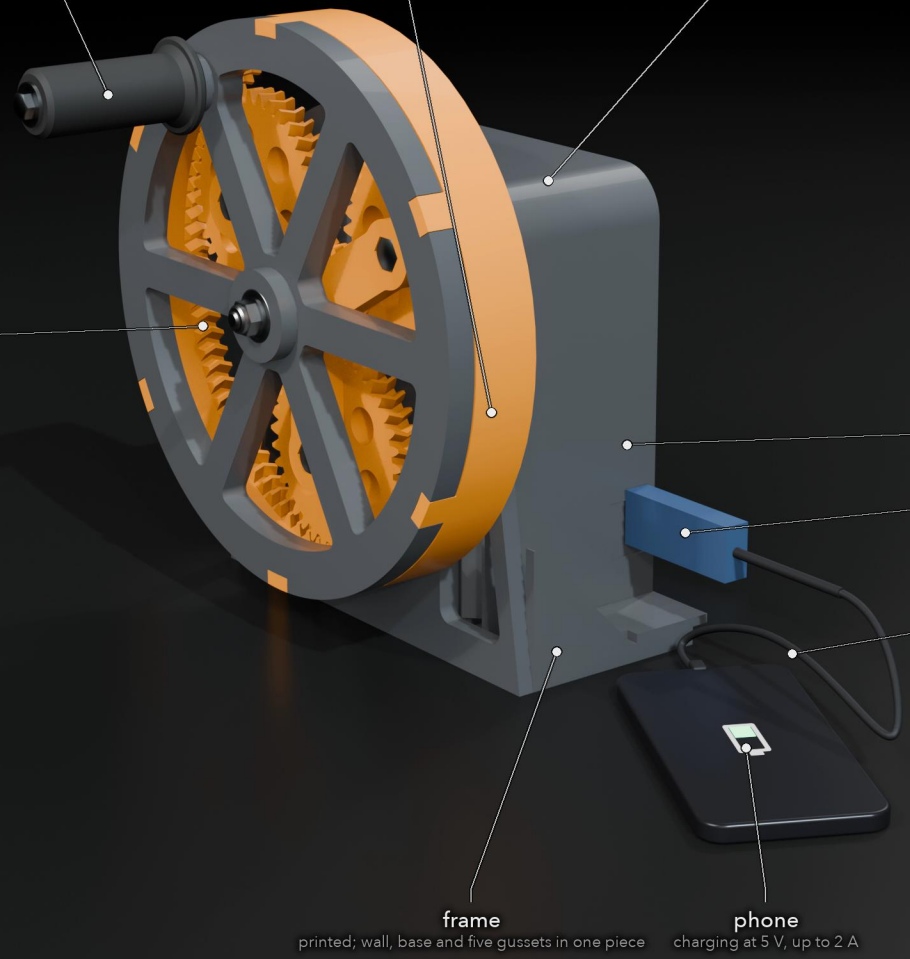
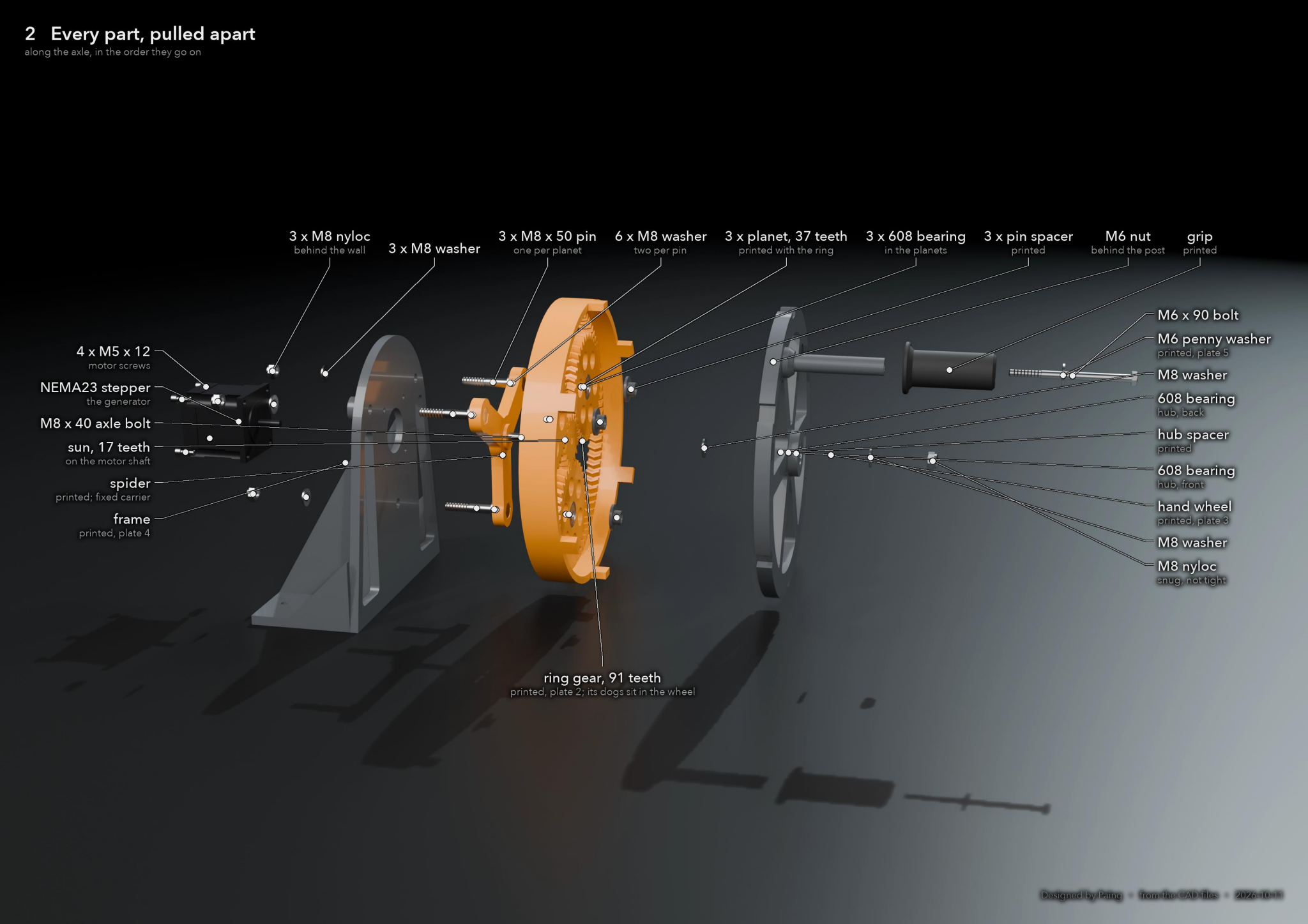
1 The machine

hand-crank phone charger, as built



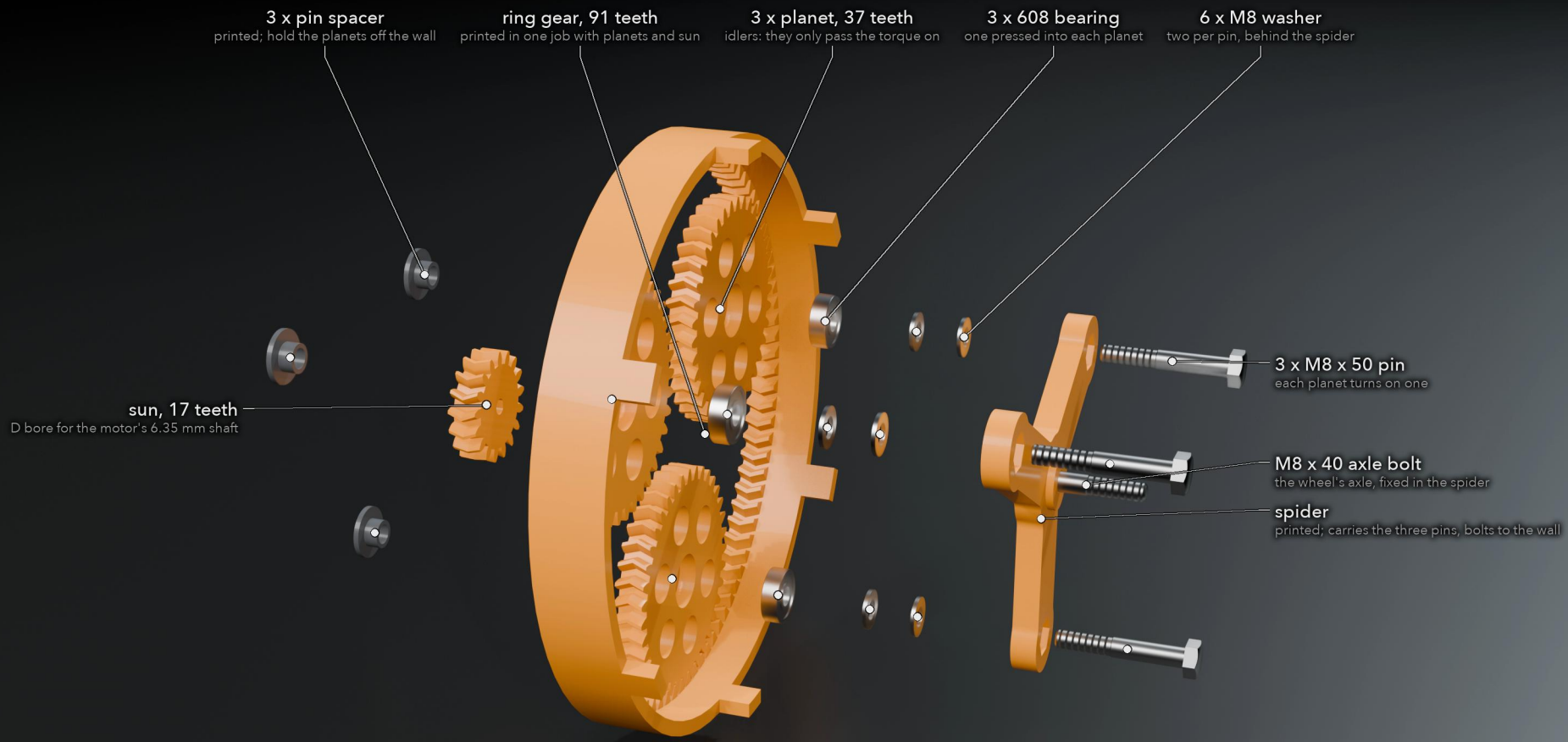
2 Every part, pulled apart

along the axle, in the order they go on



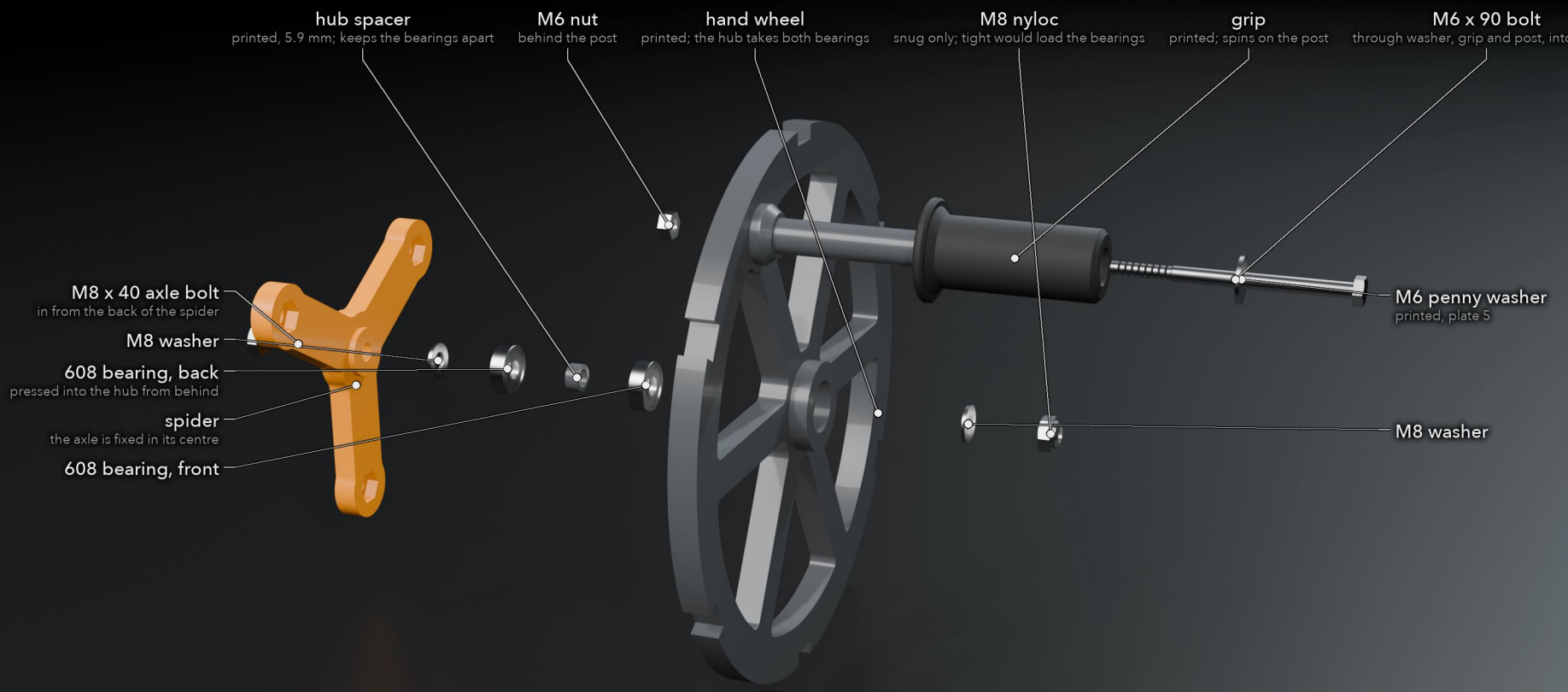
3 The gear cartridge

one planetary stage, 91 : 17 = 5.35 to 1



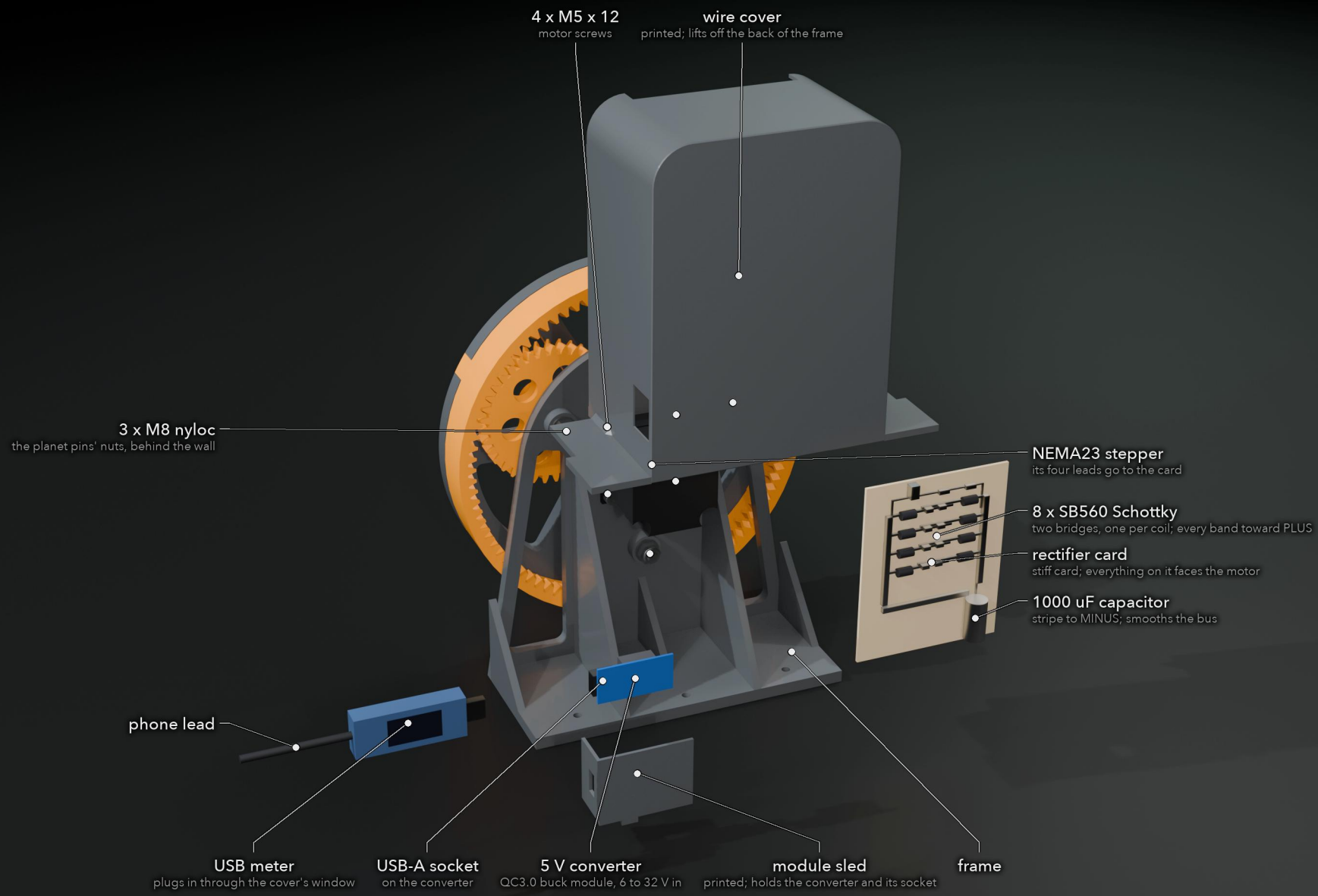
4 Hub and handle

the wheel's bearings and the grip



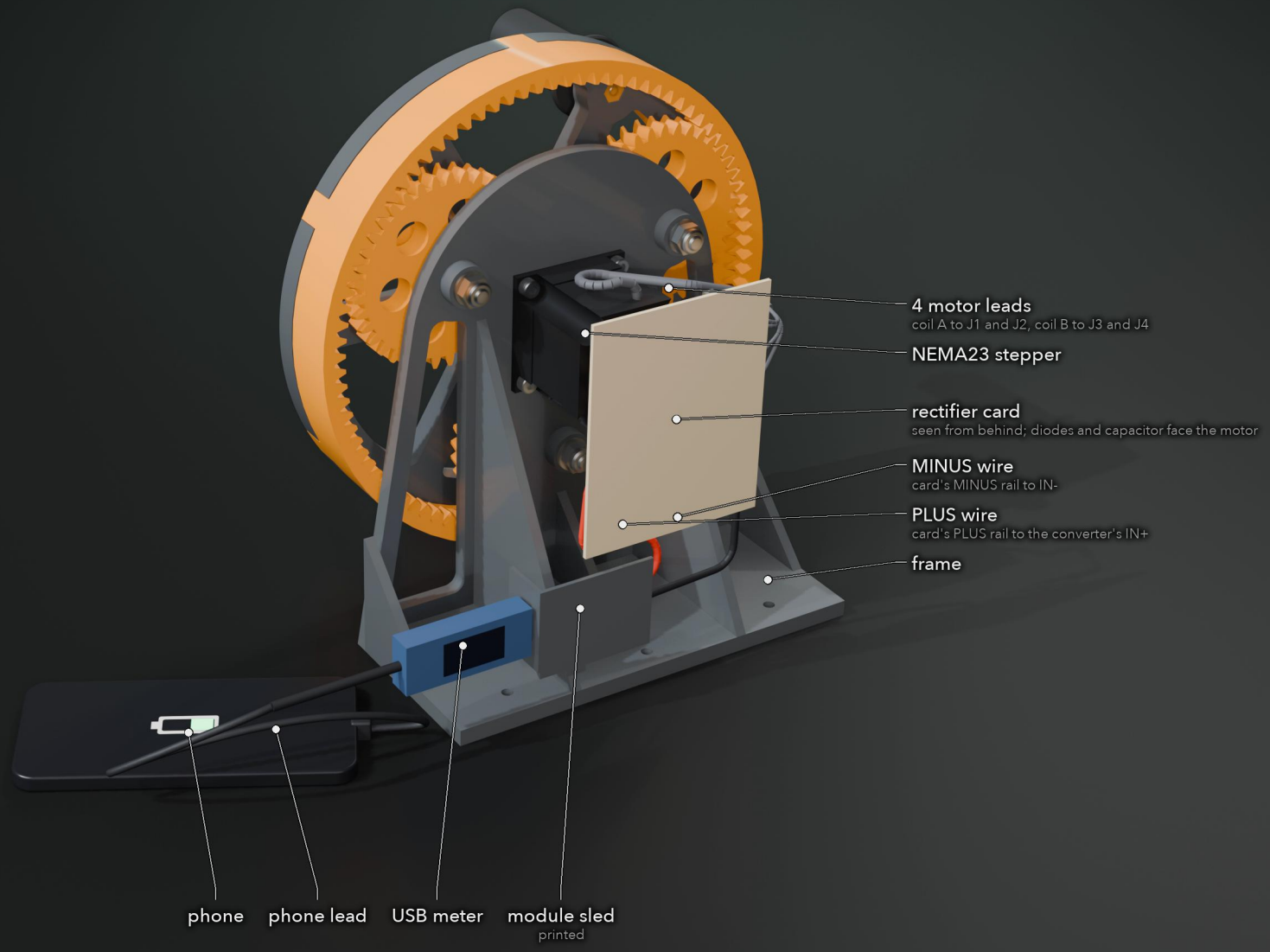
5 The electronics

lifted out of the cover



6 Electronics in place

cover off, everything wired



7 The print files

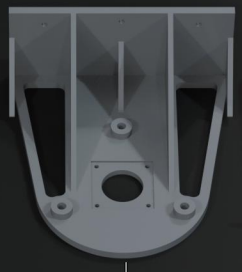
the machine's own parts, as exported, in print orientation



gear set, print in place
plate 2: ring 91, three planets 37, sun 17, teeth on the bed



hand wheel
plate 3: back on the bed, post up



frame
plate 4: front face on the bed



wire cover shell
cover file 2: open side up



spider
plate 1



12 M8 + 1 M6 penny washer
plate 5, 1.6 mm



grip
plate 1: stands on its disc



3 x pin spacer
plate 1



hub spacer
plate 1